Bayesian networks – assignment part 3

The goal of this part of the assignment is to implement the Expectation-Maximization algorithm and test it on a data set generated from the Endorisk model. You will use this data set to learn the parameters of the model. Keep a log (textfile) of the steps your algorithm does on a specific network and data set and explicate the important steps in the algorithm (e.g., convergence tests). Run the algorithm with at least 5 different random initializations of the CPTs involving hidden variables. Finally, compare the learned network with the original Endorisk model

This final part of the programming assignment complements the total assignment:

- In **1a** we asked you to implement multi-dimensional factors and do the necessary calculations on them (reduction, product, marginalization)
- In **1b** you've built on this to implement variable elimination for inference.
- In **2a** you implemented the maximalization calculation on factors
- In **2b** you have complemented the variable elimination algorithm with MAP queries
- Now, in part **3** you will add an implementation of the EM algorithm. This may be a separate executable or part of the earlier application.

The adjusted Excel sheet for part 3 will give you an indication of the weight of various components of the assessment. **An extra bonus point** if offered if you add code to save the learned network in .bif format.

Incorporate formative feedback

Please incorporate the formative feedback on the first part of the assignment in the remainder of the assignment. This feedback should give you sufficient pointers for improvement where needed for a passing grade. For the final submission (parts 1, 2, and 3 combined), please indicate clearly how and where you made improvements based on the formative feedback. For several parts (e.g. readability, testing) an overall grade is given – this can be lower than the formative feedback when the overall quality decreases!

Submission

Please submit your complete code and report for all three parts (normal VE, MAP, and EM learning) together with a programming **report** and a **video** where you demonstrate the working of your application. An example report is given in the assignments folder. For the video, you are expected to 1) explain your code, step by step, and 2) run it on your machine, to show us that it does run and shows the same results as depicted in your report. With submitting the assignment for grading you state that no unauthorized use was made of generative AI (see the course manual for more information).